MIL-F-48425A (AR)
11 SEPTEMBER 1980
SUPERSEDING
MIL-F-48425 (MU)
28 September 1973

MILITARY SPECIFICATION

FIN ASSEMBLY, M24

This specification is approved for use by the US Army Armament Research and Development Command, an is available for use by all Departments and Agencies for the Department of Defense.

1. SCOPE

1.1 This specification covers Fin Assembly, M24, for use on 81MM Mortar Ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-H-6088 - Heat Treatment of Aluminum Alloys
MIL-A-48078 - Ammunition Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1167 - Ammunition Data Cards

FSC: 1310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9268007 - Packing and Marking for Box Fiberboard,
11726889 - Fin Assembly, M24

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

ANSI B 46 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

ASTM Method E-8 - Tension Testing of Metallic Materials
ASTM Method E-18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM Method E-340 - Macroetching Metals and Alloys

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technial groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 General.- Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies.- The components and assemblies shall comply with all requirements specified on drawing (dwg.) 11726889 and associated drawings, and with all requirements specified in applicable specifications and standards.

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3.3 Heat treatment.- Heat treatment equipment, procedures and controls shall be in accordance with MIL-H-6088. The cartridge housing forging or bar stock shall be solution heat treated in such a manner as to avoid overheating of the material or cracking during quenching.

3.4 Hydrostatic test.- The fin housing shall be able to withstand a hydrostatic load of 17250 ± 250 P.S.I. when tested in accordance with 4.5.4. Each fin housing that successfully meets the requirements of this test shall be permanently stamped with the letter "H" .125 inch maximum height by .01 inch maximum depth on the outside taper surface 0.7 inch from the front face.

3.5 Data requirements.- The contractor shall generate data in accordance with the requirements of the data item descriptions cited herein and in 4.5.3.4, 4.5.4. Ammunition data cards are required (see section 6.8). All drawings and revisions of components and assemblies shall be listed on the data card.

3.6 First Article Inspection.- This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship

3.7.1 Metal characteristics.- All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, pipe type defects, inclusions, porosity, or any similar characteristics.

3.7.2 Threads.- Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawings.

3.7.3 Burrs.- No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.7.4 Foreign matter.- All parts and assemblies shall be free of chips, dirt, grease and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.- Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.- The following types of inspections shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.- The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part description	Drawings	Quantity
Fin Assembly, M24	11726889	10
Fin	11726683	10
Housing	11726795	10
Housing, prior to drilling of flash holes	11726795	10

In addition, two (2) extrusion blanks, i.e., extruded and thermally treated material stock suitable and ready for machining (See Table 1, Note d) shall be submitted from both housing stock and fin stock (total 4 extrusion blanks).

Two extrusion blanks, one selected from the front of an extrusion bar and one from the rear of an extrusion bar shall be submitted from both housing stock and fin stock (total 4 extrusion blanks). Each blank shall be identified as to location.

4.3.2 Inspections to be Performed.- See MIL-A-48078 and Table 1 specified herein.

4.3.3 Rejection.- See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48425A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Fin Assembly, M24 (dwg. 11726889)				
	Examination for defects	10		3.2	4.4.2.6
	Fin (dwg. 11726683)				
	Examination for defects	10		3.2	4.4.2.5
	Housing (dwg. 11726795)				
	Examination for defects	10		3.2	4.4.2.4
	Housing (prior to drilling flash holes) (dwg. 11726795)				
	Hydrotest	10		3.4	4.5.4
	Hydrotest, to destruction	10(a)		3.2	4.5.4
	Extrusion Blanks (front and rear blank)				
	Macroetch examination	4(b)		3.7.1	4.5.2
	Extrusion Blanks (housing)				
	Hardness test	2(c)		3.1	4.5.3.1
	Tension test	2		3.1	4.5.3.2
	Microstructure examination	1		3.3	4.5.3.3
	Extrusion Blanks (Fin)				
	Tension test	2(d)		3.1	4.5.3.2

NOTES:
(a) Test for information only. The same samples shall be used for both hydrostatic tests.
(b) Two blanks each from extrusion stock from both the fin and housing stock.
(c) Two blanks from extrusion stock fully heat treated and ready for machining.
(d) Blanks shall be of sufficient length to allow a standard flat sub-size specimen to be cut from stock.

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4.4 Quality Conformance Inspection

4.4.1 Inspection lot formation.- Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and the following as applicable:

4.4.1.1 Cartridge housing lot.- A cartridge housing lot shall consist of blanks from the same heat treatment lot of the same alloy that have been heat treated in the same heat treating equipment at the same temperature in one unchanged process. The maximum number of blanks in a heat treatment lot shall consist of 15,000 or fraction thereof. A heat treatment lot may consist of more than one solution heat treatment batch, however, the sample for hardness, tensile and microstructure tests hereinafter prescribed must be selected from and represent each solution heat treatment batch. A solution heat treatment batch shall consist of one lot of the same alloy that has been thermally treated by the same method in the same furnace equipment at the same time and temperature in one unchanged process. The identification and segregation of each solution heat treatment batch is not required after acceptable compliance with the hardness, tensile, and microstructure test, however, there shall be no more than three solution heat treatment lots in the final cartridge housing heat treatment lot.

4.4.1.2 Fin stock lot.- The extrusion lot shall consist of not more than 1000 pounds of extruded stock of the same alloy that has been thermally treated by the same method in the same furnace equipment at the same temperature and time in one unchanged process regardless of the solution heat treating method utilized. Each extrusion lot representing material that has been solution heat treated after extrusion, must be solution heat treated at the same time in the same furnace using the same temperature and time in that furnace. The extrusion lot thus constituted shall be the basis for selecting the sample blanks for the hardness and tension test hereinafter prescribed.

4.4.2 Examination.- See MIL-A-48078.

a. Sampling plans.- Sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspecton Level II.

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b. Tool control dimensions.- Dimensions marked tool control shall be inspected as follows: select and gage one fin and one housing from each machine in operation every two hours or fraction thereof and whenever tooling is changed. Prior to any tooling change, the last housing produced from the old tooling shall be gaged for the requirements of the applicable drawing. All material produced prior to periodic inspection shall be rejected and screened whenever a defective tool control dimension is found and corrective action must be taken.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.1	Fin and Housing Extrusion Bars	1	1	NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special a	Macroetch examination	(a)		3.7.1	4.5.2
Major	None defined				
Minor	None defined				

NOTES: (a) See paragraph 4.5.2 for sampling procedures. The test shall be performed on each extrusion bar.

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Fin and Housing Extrusion Blanks				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Hardness test	80		3.1	4.5.3.1
b	Tension test	(a)		3.1	4.5.3.2
c	Microstructure test	(b)		3.3	4.5.3.3
Major	None defined				
Minor	None defined				

NOTES:

(a) Two extrusion blanks shall be selected from the samples used for the hardness test for the housing stock or selected per 4.5.3.2.2 for the fin stock. Failure of either sample shall cause the respective lot to be rejected.

(b) One of the tension test samples from the housing sample only shall also be used for this test. Failure of this sample shall be cause for extrusion lot rejection of the housing blanks.

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CLASSIFICATION OF DEFECTS & TESTS MIL-F-48425A (AR)

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.3	Housing, Prior to Drilling Flash Holes		1	1	11726795
					NEXT HIGHER ASSEMBLY
					11726889
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Hydrotest		100%	3.4	4.5.4
Major	None defined				
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48425A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 11726795
4.4.2.4	Housing				NEXT HIGHER ASSEMBLY 11726889
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Flash hole(s) missing		100%	3.2	4.5.5.1
2	Hydrostatic test stamp missing		100%	3.4	Visual
Special					
a	Forward thread minor diameter oversize		100%	3.2	Gage
b	Presence of burr or foreign matter in ignition cartridge cavity or any flash hole		100%	3.7.3/ 3.7.4	Visual
c	Metal defective		100%	3.7.1	Visual
Major					
101	Diameter of housing rear of taper		0.40%	3.2	Gage
102	Diameter of knurl		0.40%	3.2	Gage
103	Diameter rear of knurl		0.40%	3.2	Gage
104	Depth to bottom of body thread cavity		0.40%	3.2	Gage
105	Depth to bottom of cartridge cavity		0.40%	3.2	Gage
106	Length from front face to inside basic diameter		0.40%	3.2	Gage
107	Length from front face to forward outside basic diameter		0.40%	3.2	Gage
108	Length from front face to rear outside basic diameter		0.40%	3.2	Gage
109	Location of flash holes		0.40%	3.2	Gage
110	Length from rear face to fin seat		0.40%	3.2	Gage
111	Length of taper		0.40%	3.2	Gage
112	Length total		0.40%	3.2	Gage
113	Length of forward thread, min.		0.40%	3.2	Gage
114	Length of rear thread, min.		0.40%	3.2	Gage

NOTES:

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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 11726795
4.4.2.4	Housing				NEXT HIGHER ASSEMBLY 11726889
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
115	Runout of pitch diameter of rear thread with pitch diameter of forward thread and front and rear faces		0.40%	3.2	Gage
116	Flash holes excessive in number		0.40%	3.2	4.5.5.2
117	Runout of taper with pitch diameter of forward thread and front face		0.40%	3.2	Gage
118	Runout of housing with pitch diameter of forward thread and front face		0.40%	3.2	Gage
Minor					
201	Depth of taper forward of body thread		0.65%	3.2	Gage
202	Radii or chamfers missing or incorrect		0.65%	3.2	Visual (See 6.6)
203	Surface finish improper		0.65%	3.7.1	Visual (See 6.7)
204	Foreign matter of burrs		0.65%	3.7.3/ 3.7.4	Visual
205	Length of rear thread, max.		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS MIL-F-48425A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11726683
4.4.2.5	Fin				NEXT HIGHER ASSEMBLY 11726889
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective		100%	3.1	Visual
Major					
101	Runout of inside diameter with outside hub diameter		0.40%	3.2	Gage
102	Diameter of fin blades (front end)		0.40%	3.2	Gage
103	Diameter of fin blades (rear end)		0.40%	3.2	Gage
104	Diameter of hub (inside)		0.40%	3.2	Gage
105	Diameter of hub (outside)		0.40%	3.2	Gage
106	Length from front face to rear face of hub		0.40%	3.2	Gage
107	Location of basic diameter of fin blades		0.40%	3.2	Gage
108	Fin blades canted in improper direction		0.40%	3.2	Visual
Minor					
201	Angle at front of fin blades		0.65%	3.2	Gage
202	Length, total		0.65%	3.2	Gage
203	Thickness of fin blades		0.65%	3.2	Gage
204	Diameter of counterbore at rear of blades		0.65%	3.2	Gage
205	Radii or chamfers missing or incorrect		0.65%	3.2	Visual (see 6.6)
206	Surface finish improper		0.65%	3.2	Visual (see 6.7)
207	Presence of burr		0.65%	3.7.3	Visual
208	Foreign matter		0.65%	3.7.4	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.6	Fin Assembly				11726889
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective		100%	3.1	Visual
Major					
101	True position of fin blades with pitch diameter of forward thread and front face		0.40%	3.2	Gage
102	True position of cartridge cavity with pitch diameter of rear thread for min. length (a)		0.40%	3.2	Gage
103	Horizontal and vertical total indicator readings of fin blades with pitch diameter of forward thread and front face		0.40%	3.2	Gage
104	Pitch diameter of forward thread, max. (a)		0.40%	3.2	Gage
105	Pitch diameter of forward thread, min. (a)		0.40%	3.2	Gage
106	Pitch diameter of rear thread, max. (a)		0.40%	3.2	Gage
107	Pitch diameter of rear thread, min. (a)		0.40%	3.2	Gage
108	Minor diameter of rear thread, max. (a)		0.40%	3.2	Gage
109	Diameter of cartridge cavity (a)		0.40%	3.2	Gage
110	Height of fin blade cant (b)		0.40%	3.2	Gage
111	Gap between fin and housing		0.40%	3.2	Visual (see 6.6)
112	Length total, max.		0.40%	3.2	Gage
113	Fin blade(s) distorted or damaged		0.40%	3.2	Visual
114	Threads damaged		0.40%	3.7.2	Visual

NOTES:
(a) Inspection to be performed on assembly, the requirement is on drawing 11726795.
(b) Inspection to be performed on assembly, the requirement is on drawing 11726683.

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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 11726889
4.4.2.6	Fin Assembly				NEXT HIGHER ASSEMBLY ----------
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Marking missing, incorrect or illegible		0.65%	3.2	Visual
202	Presence of burr		0.65%	3.7.3	Visual
203	Foreign matter		0.65%	3.7.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS MIL-F-48425A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Box, Fiberboard Prior to Sealing				9268007
					NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Tray missing or damaged		0.65%	3.2	Visual
202	Improperly packed		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Box, Fiberboard After Sealing				9268007
					NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Contents loose		0.65%	3.2	Manual
202	Box damaged (cut or damaged through all layers)		0.65%	3.2	Visual
203	Box improperly sealed		0.65%	3.2	Visual
204	Marking missing, incorrect or illegible		0.65%	3.2	Visual

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4.4.3 Testing.- Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment.- The inspection equipment required to perform the examinations and tests prescribed in this specification is identified either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the Quality Conformance Inspection tables herein. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Materials, components and processes.- Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections.- The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Macroetch examination.- A full section macroetch specimen shall be cut from the front end and the back end of the bar representing the first and last extruded material, respectively, from each individual extrusion billet. Failure to comply with the requirements of 3.7.1 will require an additional length of discard from the defective extrusion end(s) and testing of an additional macroetch specimen(s) taken from the end(s) of the remaining length of extruded material. This procedure must be continued until a sufficient length of defective extrusion has been discarded and the requirements of 3.7.1 are met. Suitable identification of each length of extrusion and corresponding macroetch must be maintained until the extruded material is accepted. The specimen shall be macroetched using the criteria prescribed in ASTM Method E-340 except that sodium hydroxide shall be used as the macroetching reagent.

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4.5.3 Mechanical properties

4.5.3.1 Hardness

4.5.3.1.1 Cartridge housing or fin stock lot.- Prior to finish machining but after all heat treatment, 80 blanks representing each solution heat treatment lot (as defined by 4.4.1) shall be subjected to a hardness test on either end of the cartridge housing or the approximate center of the cross section of the blank as applicable. The test shall be performed using the equipment and procedures prescribed by ASTM Method E-18, B scale. The hardness values shall be recorded and the two blanks representing the hardest and softest material shall be set aside for the tension test.

4.5.3.2 Tension test

4.5.3.2.1 Cartridge housing lot.- Two cartridge housings, one of the hardest and one of the softest as determined by 4.5.3.1.1 representing each solution heat treatment lot shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. From each of the housings selected, one tension test specimen conforming to the largest obtainable standard round type specimen as prescribed by ASTM Method E-8 shall be taken with the approval of the procuring agency, the Government shall not be restricted from testing additional specimens at other locations to determine that the mechanical properties conform to drawing requirements. The criteria for performing this test shall be as prescribed by ASTM Method E-8.

4.5.3.2.2 Fin stock extrusion lots.- From the fin stock lot (as defined by 4.4.1.2), one fin from the first 1000 pounds or fraction thereof shall be subjected to test after all heat treatment has been completed. One additional fin from each 2000 pounds or fraction thereof for the remainder of the extrusion lot, shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirement of the applicable drawing shall be cause for rejection of the lot. The test criteria shall be as prescribed by ASTM Method E-8 and each test specimen shall be of a subsize flat type taken longitudinally from a fin blade. The Government shall not be restricted from testing additional specimens at other locations to determine that the mechanical properties meet the drawing requirements for yield strength and elongation.

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4.5.3.3 Microstructure examination.- From the specimens submitted for tension testing in 4.5.3.2.1, one wafer, approximately one-half inch in length shall be cut longitudinally and prepared for metallographic examination. When so prepared, the specimen shall be etched with Keller's reagent and examined by scanning under a microscope at a minimum magnification of 500 diameters. Overheating shall be evidenced by the presence of rosettes in the microstructure or incipient melting at the grain boundaries (see 6.5). Evidence of the above shall cause rejection of the representative heat treatment or bar stock lot.

4.5.3.4 Quality inspection report.- The contractor shall submit a quality inspection test, demonstration, and evaluation report in accordance with Data Item Description DI-R-1724 (tailored) giving the detailed test results from the above tests (4.5.3.1, 4.5.3.2, and 4.5.3.3).

4.5.4 Hydrostatic test.- After final machining but prior to drilling flash holes and prior to gaging, each cartridge housing shall be subjected to a hydrostatic test to determine compliance with the requirements of 3.4 and 3.7.1. Failure to meet the requirements of 3.4 and 3.7.1 shall be cause for rejection of the housing. The housing shall be tested using equipment capable of performing the test accurately and consistently. In performing the test, the pressure will be applied gradually until the required minimum pressure is reached and held for three seconds. The section where the fin is assembled to the housing shall be fully supported on the outside diameter during test. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 4 hours of continuous operation. It shall also be calibrated prior to restarting the operation after a lay-off of three hours or more. If it has been found that the equipment is out of calibration in such a direction that the pressure was actually lower than required by the applicable drawing, the fins tested since the last satisfactory calibration shall be retested after the calibration error has been corrected. Rupture, cracking, or bulging shall be evidence of failure and cause for rejection of the housing. Each housing which has successfully passed the test shall be stamped in accordance with the applicable drawing. For first article inspection only, the test pressure shall be increased until the housing ruptures and the maximum pressure shall be recorded for information purposes. The contractor shall submit a monthly quality defect report in accordance with Data Item Description DI-R-1721 summarizing the defects found.

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4.5.5 Flash hole inspection

4.5.5.1 Missing flash holes.- Each fin housing shall be inspected by an electronic and/or mechanical system capable of determining the presence of the required number of flash holes (see drawing).

4.5.5.2 Excessive flash holes.- Each fin housing shall be inspected by a method capable of determining the presence of flash holes in excess of the drawing requirement.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be in accordance with the requirements of dwg. 9268007.

6. NOTES

6.1 Intended use.- The components covered by this specification are intended for use on the mortar, 81MM, HE, M374A3, and 81MM, smoke, W.P., M375A3. This is a pressure vessel application.

6.2 Ordering data.- See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals.- See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, NJ 07801.

6.4 Drawings. - Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Microstructure examination. - Microstructure examination of solution heat treat specimens should be performed by a skilled metallographer familiar with aluminum alloys.

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6.6 Visual examination qualification.- When compliance with the applicable requirements is in doubt as a result of visual examination, these characteristics may be measured or gaged to determine acceptability.

6.7 Surface roughness.- The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination. Four power (4X) magnification may be used in performing visual examination.

6.8 Data requirements. - Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.5.3.4	Quality Inspection Test, Demonstration and Evaluation Report	DI-R-1724 (tailored)
4.5.4	Quality Inspection Defect Report	DI-R-1721
3.5	Ammunition Data Card	DI-L-1410

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.9 Changes for previous issue. - Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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